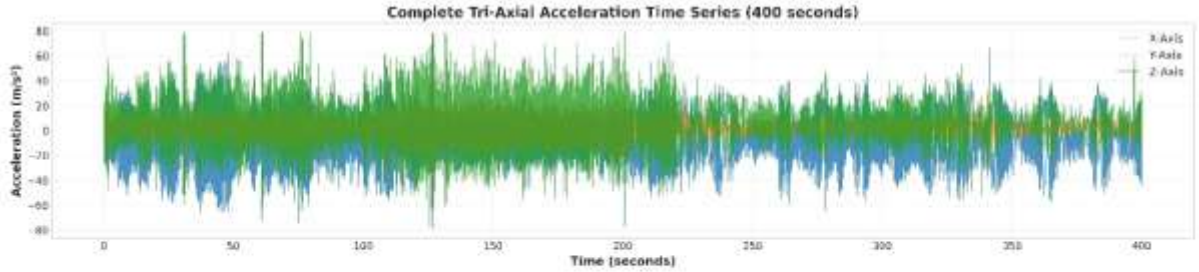
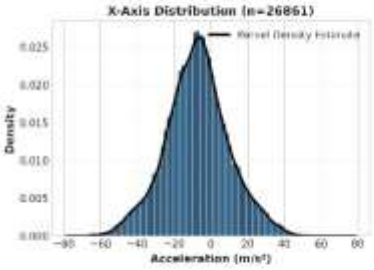
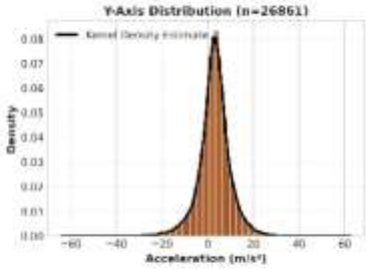
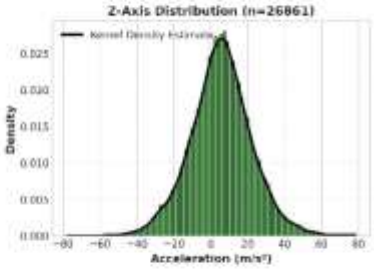
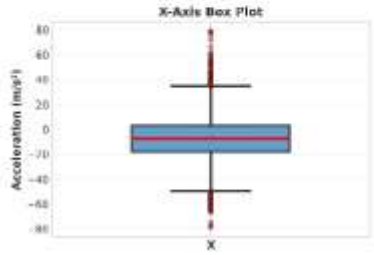
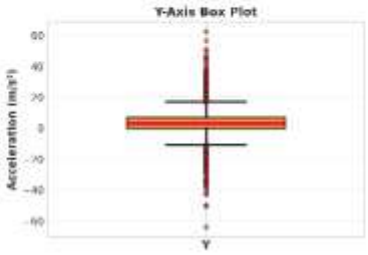

Complete Tri-Axial Acceleration Time Series (400 seconds)
Acceleration (m/s²)
Time (seconds)
X-Axis
Y-Axis
Z-Axis

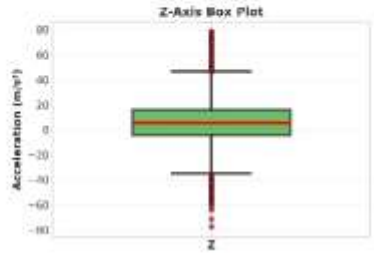

X-Axis Distribution (n=26861)
Kernel Density Estimate
Density
Acceleration (m/s²)

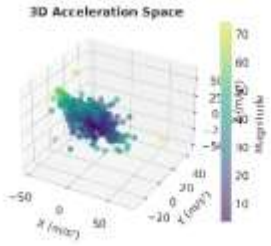

Y-Axis Distribution (n=26861)
Kernel Density Estimate
Density
Acceleration (m/s²)

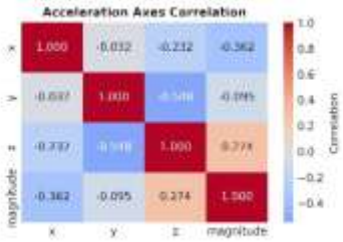

Z-Axis Distribution (n=26861)
Kernel Density Estimate
Density
Acceleration (m/s²)

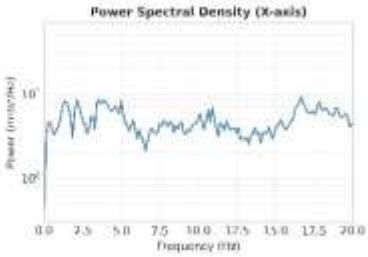

X-Axis Box Plot
Acceleration (m/s²)
X

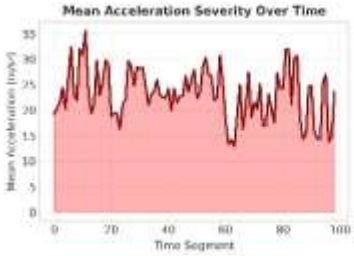

Y-Axis Box Plot
Acceleration (m/s²)
Y

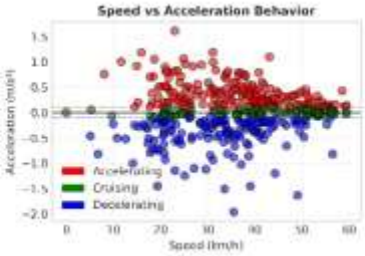

Z-Axis Box Plot
Acceleration (m/s²)
Z


3D Acceleration Space
magnitude

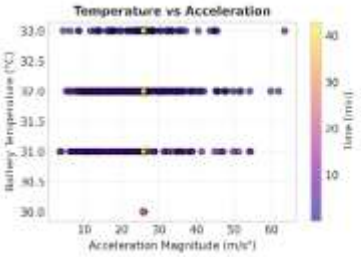

Acceleration Axes Correlation
x
y
z
magnitude
1.000
-0.032
-0.232
-0.362
-0.032
1.000
-0.548
-0.095
-0.232
-0.548
1.000
0.274
-0.362
-0.095
0.274
1.000
Correlation

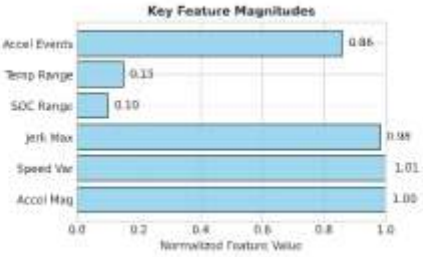

Power Spectral Density (X-axis)
Frequency (Hz)

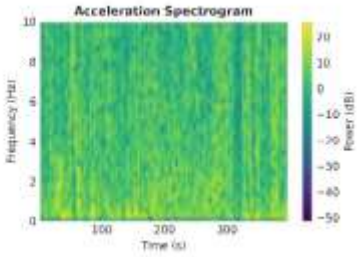

Mean Acceleration Severity Over Time
Mean Acceleration (m/s²)
Time Segment

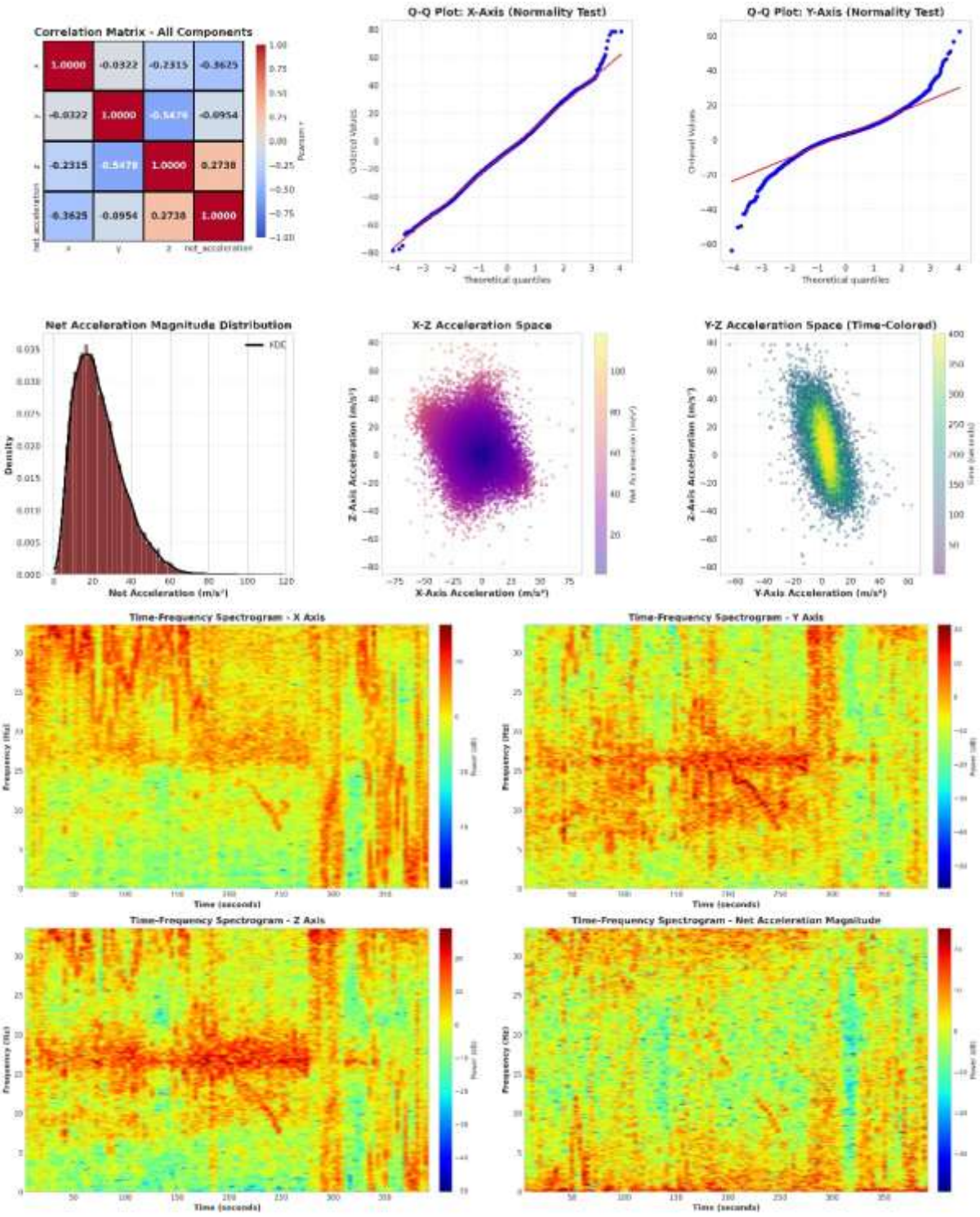

Speed vs Acceleration Behavior
Acceleration (m/s²)
Accelerating
Cruising
Decelerating
Speed (km/h)

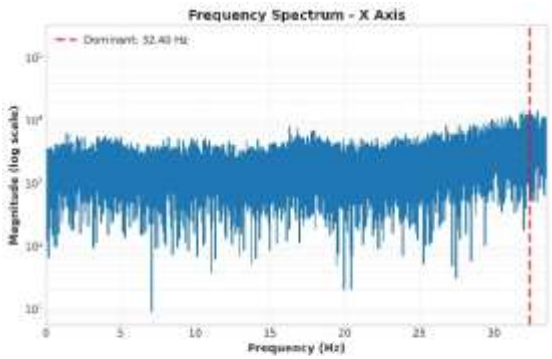
Battery Discharge vs Driving Intensity
Battery SOC (%)
Time Segment

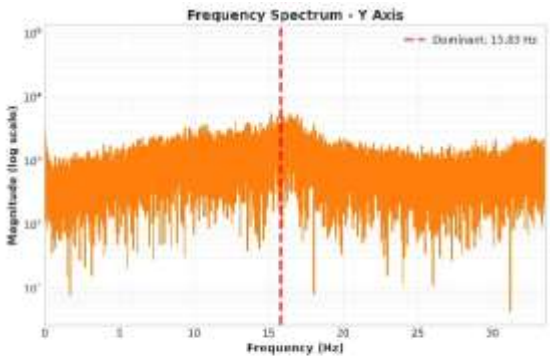

Temperature vs Acceleration
Battery Temperature (°C)
Acceleration Magnitude (m/s²)
Time (min)

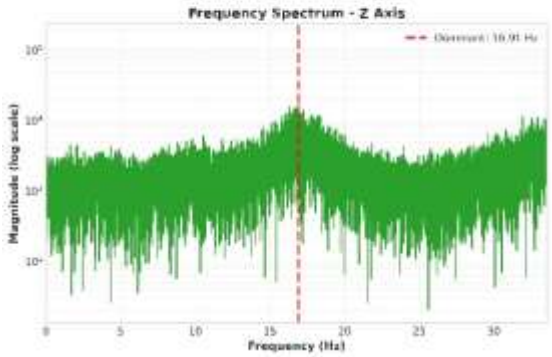

Key Feature Magnitudes
Accel Events
Temp Range
SOC Range
Jerk Max
Speed Var
Accel Mag
0.86
0.13
0.10
0.98
1.01
1.00
Normalized Feature Value

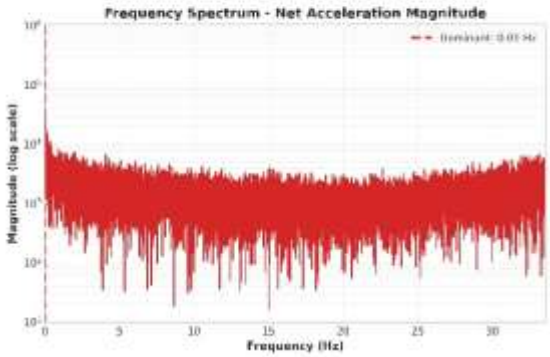

Acceleration Spectrogram
Frequency (Hz)
Time (s)
Power (dB)

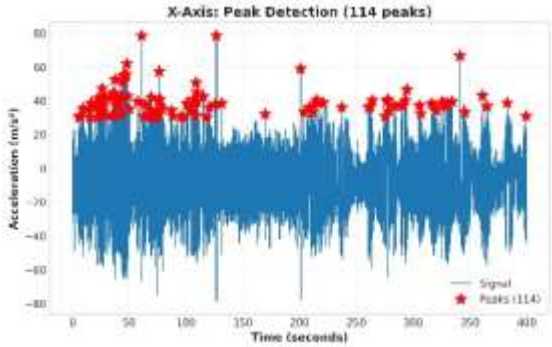

Correlation Matrix - All Components
1.0000
-0.0322
-0.2315
-0.3625
-0.0322
1.0000
-0.5478
-0.0954
-0.2315
0.5478
1.0000
0.2738
-0.3625
-0.0954
0.2738
1.0000
x
y
z
net_acceleration
Pearson r
Q-Q Plot: X-Axis (Normality Test)
Q-Q Plot: Y-Axis (Normality Test)
Ordered Values
Theoretical quantiles
Net Acceleration Magnitude Distribution
KDE
Density
Net Acceleration (m/s²)
X-Z Acceleration Space
Z-Axis Acceleration (m/s²)
X-Axis Acceleration (m/s²)
Y-Z Acceleration Space (Time-Colored)
Y-Axis Acceleration (m/s²)
Time-Frequency Spectrogram - X Axis
Time-Frequency Spectrogram - Y Axis
Time-Frequency Spectrogram - Z Axis
Time-Frequency Spectrogram - Net Acceleration Magnitude
Frequency (Hz)
Time (seconds)
Power (dB)

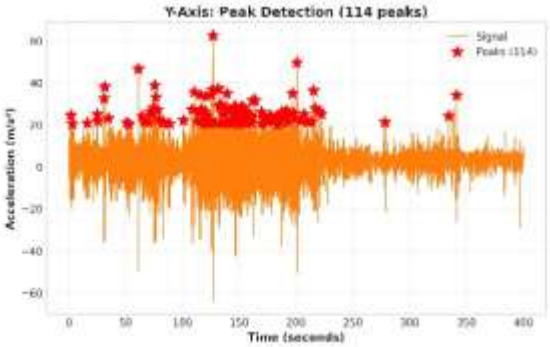

Frequency Spectrum - X Axis
Dominant: 32.40 Hz
Magnitude (log scale)
Frequency (Hz)

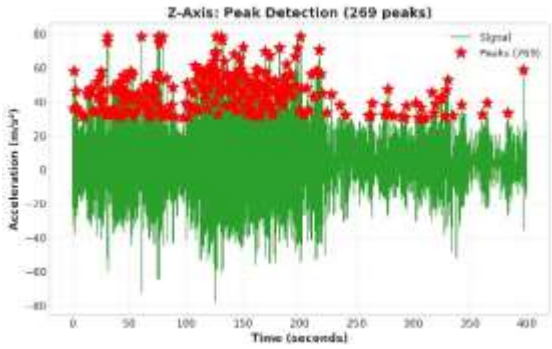

Frequency Spectrum - Y Axis
Dominant: 15.83 Hz
Magnitude (log scale)
Frequency (Hz)

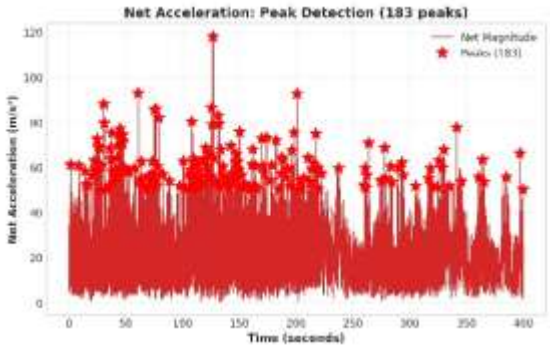

Frequency Spectrum - Z Axis
Dominant: 16.91 Hz
Magnitude (log scale)
Frequency (Hz)


Frequency Spectrum - Net Acceleration Magnitude
Dominant: 0.01 Hz
Magnitude (log scale)
Frequency (Hz)


X-Axis: Peak Detection (114 peaks)
Signal
Peaks (114)
Acceleration (m/s²)
Time (seconds)


Y-Axis: Peak Detection (114 peaks)
Signal
Peaks (114)
Acceleration (m/s²)
Time (seconds)


Z-Axis: Peak Detection (269 peaks)
Signal
Peaks (269)
Acceleration (m/s²)
Time (seconds)


Net Acceleration: Peak Detection (183 peaks)
Net Magnitude
Peaks (183)
Net Acceleration (m/s²)
Time (seconds)

ACCELEROMETER DATA: COMPREHENSIVE ANALYSIS DASHBOARD

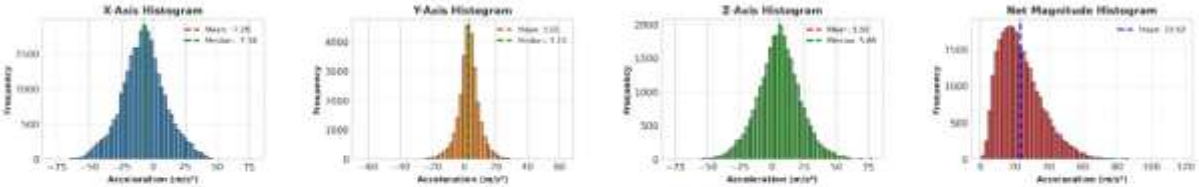
Statistical Summary

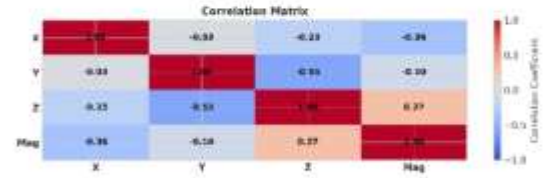
Metric	X-Axis	Y-Axis	Z-Axis	Magnitude
Mean	7.28	5.01	5.59	23.52
Std Dev	17.18	9.83	16.69	12.64
Min	-78.45	-61.50	-77.77	0.51
Max	78.46	62.30	78.38	116.14

DATASET INFORMATION

Total Samples: 26,861
Duration: 399.81 sec
Sampling Rate: 67.18 Hz
Nyquist Freq: 33.59 Hz

TIME SPAN
Start: 0.20 s
End: 400.01 s
Interval: 0.0149 s



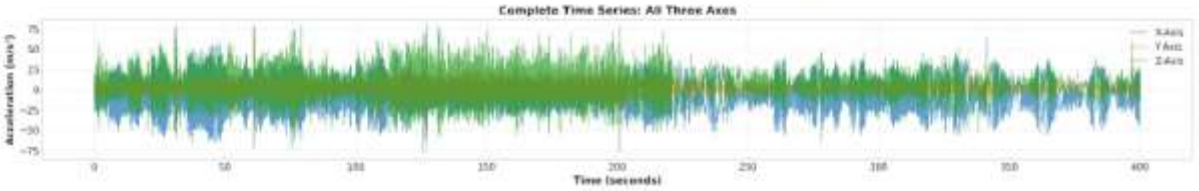


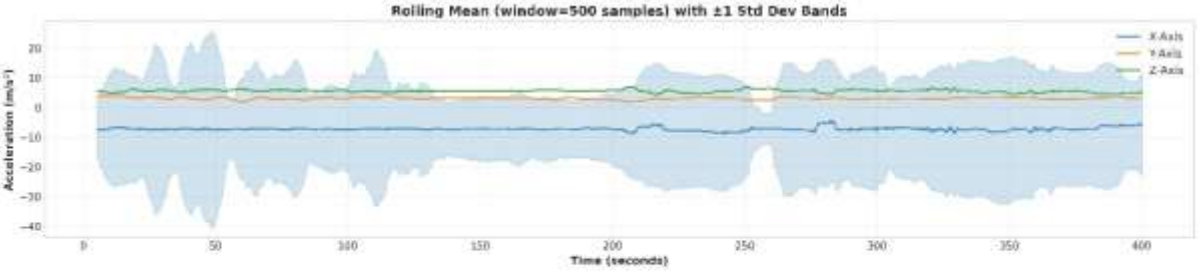
ENERGY & FREQUENCY ANALYSIS

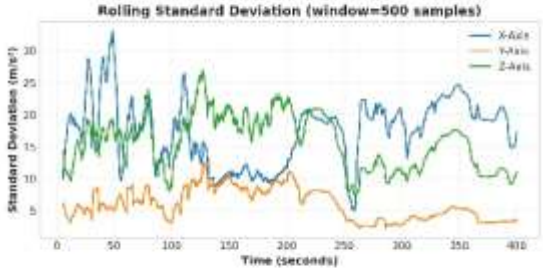
RMS Values (m/s²):
X-Axis: 18.0378
Y-Axis: 7.4842
Z-Axis: 17.5897
Total: 35.4486

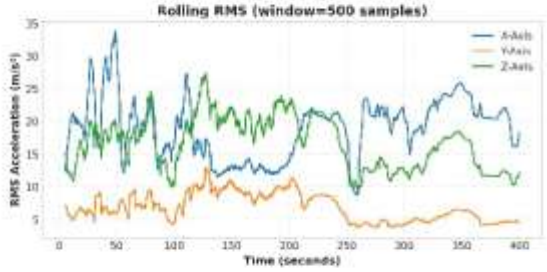
Dominant Frequencies:
X-Axis: 37.48 Hz
Y-Axis: 35.83 Hz
Z-Axis: 34.93 Hz

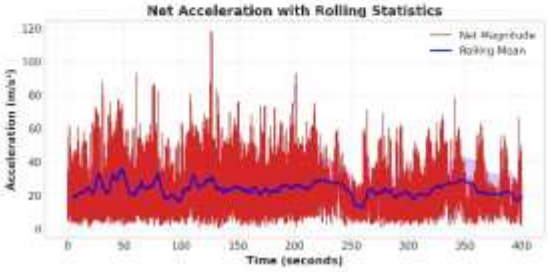
Signal Properties:
- Stationary (ADF test)
- Non-normal (Jarque-Bera)
Stationarity: Confirmed

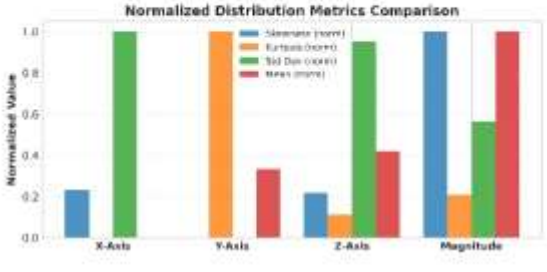


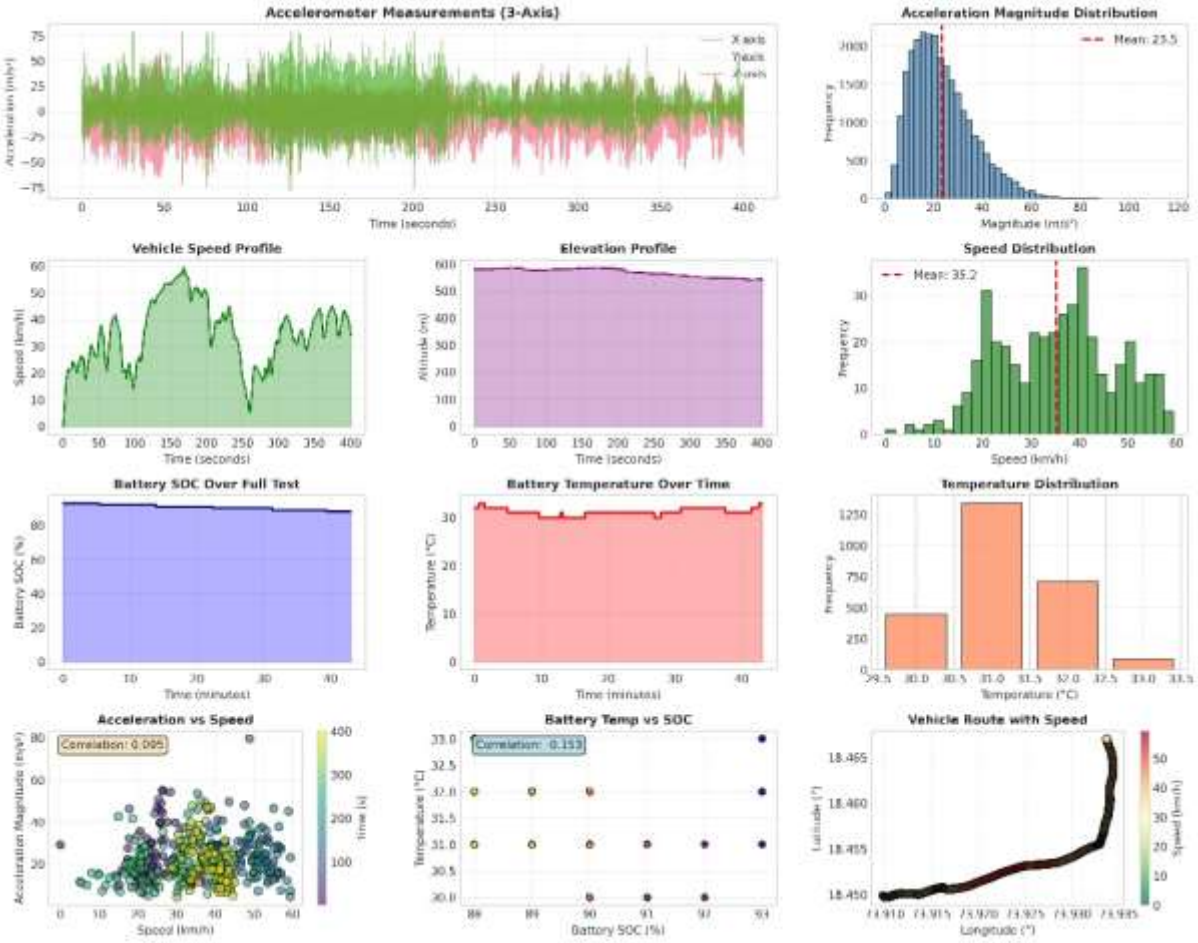











Accelerometer Measurements (3-Axis)
Acceleration (m/s²)
Time (seconds)
Acceleration Magnitude Distribution
Mean: 23.5
Frequency
Magnitude (m/s²)
Vehicle Speed Profile
Speed (km/h)
Time (seconds)
Elevation Profile
Altitude (m)
Time (seconds)
Speed Distribution
Mean: 35.2
Frequency
Speed (km/h)
Battery SOC Over Full Test
Battery SOC (%)
Time (minutes)
Battery Temperature Over Time
Temperature (°C)
Time (minutes)
Temperature Distribution
Frequency
Temperature (°C)
Acceleration vs Speed
Correlation: 0.095
Acceleration Magnitude (m/s²)
Speed (km/h)
Time (s)
Battery Temp vs SOC
Correlation: 0.153
Temperature (°C)
Battery SOC (%)
Vehicle Route with Speed
Latitude (°)
Longitude (°)
Speed (km/h)